

B - Split Ticketing

分值: 200 分

Problem Statement

There are N stations $1, 2, \dots, N$, arranged in a straight line from west to east in this order.

The AtCoder Railway train passes through these N stations and runs from west to east.

For any two integers i, j satisfying $1 \leq i < j \leq N$, the cost of boarding the train at station i and getting off at station j is $C_{i,j}$.

共有 N 个车站 $1, 2, \dots, N$, 按从西到东的顺序排列在一条直线上。AtCoder 铁路的列车会经过这 N 个车站, 从西向东行驶。对于满足 $1 \leq i < j \leq N$ 的任意两个整数 i, j , 在车站 i 上车、在车站 j 下车的费用为 $C_{i,j}$ 。

Determine whether there exist three integers a, b, c such that:

判断是否存在三个整数 a, b, c 满足以下条件:

- $1 \leq a < b < c \leq N$
 - The total cost of boarding the train at station a , getting off at station b , then boarding the train again at station b , and getting off at station c is less than the cost of boarding the train at station a and getting off at station c .
 - $1 \leq a < b < c \leq N$
 - 在车站 a 上车、在车站 b 下车, 然后在车站 b 再次上车、在车站 c 下车的总费用, 小于在车站 a 上车、在车站 c 下车的费用。
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Constraints

- $3 \leq N \leq 100$
- $1 \leq C_{i,j} \leq 10^9$
- All input values are integers.
- $3 \leq N \leq 100$
- $1 \leq C_{i,j} \leq 10^9$

- 所有输入值均为整数。

Input

The input is given from Standard Input in the following format:

输入以如下格式从标准输入给出：

N
 $C_{1,2} \ C_{1,3} \ \dots \ C_{1,N}$
 $C_{2,3} \ \dots \ C_{2,N}$
 $\vdots$
 $C_{N-1,N}$

Output

If there exist three integers a, b, c satisfying the conditions, output Yes; otherwise, output No, on a single line.

如果存在三个满足条件的整数 a, b, c , 输出 Yes; 否则输出 No, 占一行。

Sample Input 1

```
3
45 450
45
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Sample Output 1

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Yes
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Choosing $(a, b, c) = (1, 2, 3)$,
 $C_{a,b} + C_{b,c} = C_{1,2} + C_{2,3} = 45 + 45$
 $C_{a,c} = C_{1,3} = 450$
so the conditions are satisfied.

选择 $(a, b, c) = (1, 2, 3)$, $C_{a,b} + C_{b,c} = C_{1,2} + C_{2,3} = 45 + 45$ $C_{a,c} = C_{1,3} = 450$ 因此条件被满足。

Sample Input 2

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4
25 40 65
30 55
25
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Sample Output 2

No

No choice of (a, b, c) satisfies the conditions.

不存在满足条件的 (a, b, c) 的选择。